

CSE-2430 Mid Examination Spring-2025

Marks - 8

B1

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 256 MB

In the kingdom of Vectronia, brave adventurers venture into ancient dragon caves in search of legendary treasure. These caves are filled with loot gems of various values. However, not all loot is worth collecting. The royal treasury wants only the top k highest-value gems, and your task is to help the adventurers make the most efficient decision. Each cave holds a list of gem values, and your job is to extract the k largest gem values and return them in descending order for appraisal at the palace.

Input

The first line contains two integers n and k ($1 \leq k \leq n \leq 10^5$) — the number of gems and the number of gems to collect.

The second line contains n integers the value of each gem ($-10^6 \leq \text{value} \leq 10^6$).

Output

Print k integers the top k gem values in descending order.

Sample Input 1	Sample Output 1
7 3 5 -2 9 1 7 3 6	9 7 6
Sample Input 2	Sample Output 2
5 2 -1 -2 -3 -4 -5	-1 -2

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Marks - 7

B9

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 512 MB

In the land of Myrrhvale, there lies an ancient oracle who can only be summoned with sacred words that contain only vowels and read the same backward as forward.

A young coder named Lyra has come across a mysterious string carved into a glowing stone tablet. She believes it might be the oracle's word. To confirm, she must check if the string is made up of only vowels and if it is a palindrome.

Your task is to help Lyra by writing a program that checks whether a given string is a vowel-palindrome, that is:

- Every character in the string must be a vowel ('a', 'e', 'i', 'o', 'u')
- The string must read the same forward and backward

Input

A single string s consisting of lowercase English letters.

Length: $1 \leq |s| \leq 1000$

Output

Print:

"YES" if the string is a vowel-palindrome

"NO" otherwise

Sample Input 1	Sample Output 1
aeiouuoiea	YES
Sample Input 2	Sample Output 2
aeibcbaiea	NO

Marks - 5

B3

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 512 MB

Given an array `arr` of n integers, you are tasked with creating a prefix sum array `p` and then answering q queries. Each query asks for the sum of elements in a subarray from index s to t (1-based index).

Input:

- The first line contains an integer n — the size of the array.
- The second line contains n integers separated by spaces, representing the array `arr`.
- The third line contains an integer q — the number of queries () **Mallolous Rate**
- The next q lines each contain two integers s and t — the range for the query ()
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Output:

For each query, output a single integer representing the sum of the elements from index s to t .

Constraints:

- $1 \leq n, q \leq 10^5$
- $-10^5 \leq \text{arr}[i] \leq 10^5$
- $1 \leq s \leq t \leq n$

Sample Input 1

5
1 2 3 4 5
3
1 3
2 4
1 5

Sample Output 1

6
9
15